Ahmed Saeed

Assistant Professor, Georgia Institute of Technology
Also known as Ahmed Said Mohamed Tawfik Issa

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Research Interests

Theory, design, and implementation of scalable **computer networks** and **computer systems**, including resource scheduling, congestion control, and formal methods.

Education

Georgia Institute of Technology, Atlanta, Georgia, USA
PhD in Computer Science — August 2019

Georgia Institute of Technology, Atlanta, Georgia, USA
M.Sc. in Computer Science — May 2018

Alexandria University, Alexandria, Egypt
Bachelor's of Science, Computer and Systems Engineering — July 2010

Employment

Georgia Institute of Technology, Atlanta, Georgia, USA
Assistant Professor — Aug. 2021 - now

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA
Postdoctoral Associate — Sep. 2019 - Jul. 2021

Google Inc., Sunnyvale and Mountain View, CA, USA
Student Researcher — Aug. 2018 - Dec. 2018
Software Engineering Intern — May 2018 - Aug. 2018
Software Engineering Intern — May 2016 - Jan. 2017
Software Engineering Intern — May 2015 - Aug. 2015

Carnegie Mellon University Qatar and Qatar University, Doha, Qatar
Research Fellow — Nov. 2013 - July 2014

Egypt-Japan University for Science and Technology, Alexandria, Egypt
Research Assistant — July 2010 - Aug. 2013

Nile University, Cairo, Egypt
Undergraduate Research Assistant — July 2009 - Aug. 2009

Honors and Awards

NSF CAREER Award

Faculty Fellow of the Brook Byers Institute for Sustainable Systems (BBISS)

DARPA Riser, 2022
Top-five honoree at DARPA Forward at Georgia Tech; invited to DARPA Demo Day 2023 at the Pentagon.

Class of 1969 Teaching Fellow, 2022

Inclusive STEM Teaching Fellow, 2022

Google PhD Fellowship in Systems and Networking (2017-2019)

Funding

CAREER: Integrated Load and Resource Management for High-Utilization Datacenters
NSF award 2542973; PI
$595,834; 06/01/2026–05/31/2031

NeTS: Small: Understanding and Weathering Variability in Low Earth Orbit Satellite Networks
NSF award CNS-2345827; PI with Ketan Bhardwaj
$600,000; 10/2024–10/2027

Collaborative Research: CNS Core: Medium: High-performance Network Stacks for the Edge
NSF award CNS-2212098; PI with Alex Daglis and Adam Belay (MIT)
$1,198,140; GT: $748,140; 10/2022–09/2025

Collaborative Research: CNS Core: Medium: Robust Behavioral Analysis and Synthesis of Network Control Protocols Using Formal Verification
NSF award CNS-2212103; Co-PI with Hari Balakrishnan (MIT)
$1,199,020; GT: $299,020; 10/2022–09/2025

From Design to Print: An Analytical Study of Circuit Synthesis from Nanomodular Electronics
Speculative Technologies; PI with Alex Daglis
Unrestricted gift, $40,000; 08/2023

Google Research Award
Google; Unrestricted gift
$60,000; 06/2022

Public Engagement on Data Centers

Public Policy Op-Ed May 2026
"Georgia's data center boom requires a strategy, not just hoping for the best", *The Atlanta Journal-Constitution (AJC)*, with Josiah Hester.

International Radio Interview Apr. 2026
Featured on "Big data, big cata: enquête sur l'empreinte écologique des data centers", *C'est pas du vent*, Radio France Internationale (RFI).

Environmental Law Conference Talk Mar. 2026
2026 Red Clay Conference, University of Georgia School of Law; delivered "Demystifying Datacenters: AI–Energy–Sustainability."

Panelist on Datacenter Community Impact Oct. 2025
DeKalb County Data Center Town Hall; panel discussion on data-center energy and water use, infrastructure, costs, taxation, jobs, and future AI demand.

State Policy and Transparency Commentary Sep. 2025
Quoted in "How Georgia became the 'wild west' of data centers. Is transparency on the horizon?", *Columbus Ledger-Enquirer*.

Regional News Expert Commentary Aug. 2025
Quoted in "Data centers need a lot more juice. Georgia consumers fear being squeezed", *The Atlanta Journal-Constitution*.

Conference Publications

[1] **Svalinn: Overload Control in Large-Scale Servers with Multiple Resource Bottlenecks**
USENIX OSDI 2026
Bhaskar Pardeshi, Peidi Song, Ahmed Saeed

[2] **Morp4: A Dynamic Network Telescope**
USENIX NSDI 2026
Iliana Maria Xygkou, Jithin K. Sojan, Dhruv Rauthan, Feng Zhu, Thomas Holterbach, Shane Alcock, Brian Flanagan, Ahmed Saeed, Alberto Dainotti

[3] **Governing Datacenter Growth: Regulatory Challenges, Technical Constraints, and Policy Responses Across States**
TPRC 54
Ahmed Saeed, Josiah Hester

[4] **Assessing LEO Satellite Networks for National Emergency Failover**
ACM IMC 2025
Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary S. Bischof, Fabi'an E. Bustamante, Alberto Dainotti, Ahmed Saeed

[5] **CoreSync: A Protocol for Joint Core Scheduling and Overload Control of μs-Scale Tasks**
IEEE ICNP 2025
Bhaskar Pardeshi, Eric Stuhr, Ahmed Saeed

[6] **Are LEO Networks the Future of National Emergency Failover? – A Quantitative Study and Policy Blueprint**
TPRC 53
Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary S. Bischof, Fabi'an E. Bustamante, Alberto Dainotti, Ahmed Saeed

[7] **LDB: An Efficient Latency Profiling Tool for Multithreaded Applications**
USENIX NSDI 2024
Inho Cho, Seo Jin Park, Ahmed Saeed, Adam Belay, Mohammad Alizadeh

[8] **Protego: Overload Control for Applications with Unpredictable Lock Contention**
USENIX NSDI 2023
Inho Cho, Ahmed Saeed, Seo Jin Park, Mohammad Alizadeh, Adam Belay

[9] **A Characterization of Route Variability in LEO Satellite Networks**
PAM 2023
Vaibhav Bhosale, Ahmed Saeed, Ketan Bhardwaj, Ada Gavrilovska

[10] **Toward Formally Verifying Congestion Control Behavior**
ACM SIGCOMM 2021
Venkat Arun, Mina Tahmasbi Arashloo, Ahmed Saeed, Mohammad Alizadeh, Hari Balakrishnan

[11] **Throughput-Fairness Tradeoffs in Mobility Platforms**
ACM MobiSys 2021
Arjun Balasingam, Karthik Gopalakrishnan, Radhika Mittal, Venkat Arun, Ahmed Saeed, Mohammad Alizadeh, Hamsa Balakrishnan, Hari Balakrishnan

[12] **Scouting the Path to a Million-Client Server**
PAM 2021
Yimeng Zhao, Ahmed Saeed, Ellen Zegura, Mostafa Ammar

[13] **Annulus: A Dual Congestion Control Loop for Datacenter and WAN Traffic Aggregates**
ACM SIGCOMM 2020
Ahmed Saeed, Varun Gupta, Prateesh Goyal, Milad Sharif, Rong Pan, Mostafa Ammar, Ellen Zegura, Keon Jang, Mohammad Alizadeh, Abdul Kabbani, Amin Vahdat

[14] **Overload Control for μs-scale RPCs with Breakwater**
USENIX OSDI 2020
Inho Cho, Ahmed Saeed, Joshua Fried, Seo Jin Park, Mohammad Alizadeh, Adam Belay

[15] **Eiffel: Efficient and Flexible Software Packet Scheduling**
USENIX NSDI 2019
Ahmed Saeed, Yimeng Zhao, Nandita Dukkipati, Ellen Zegura, Mostafa Ammar, Khaled Harras, Amin Vahdat

[16] **zD: A Scalable Zero-Drop Network Stack at End Hosts**
ACM CoNEXT2019
Yimeng Zhao, Ahmed Saeed, Ellen Zegura, Mostafa Ammar

[17] **Unison: Enabling Content Provider/ISP Collaboration using a vSwitch Abstraction**
IEEE ICNP 2019
Yimeng Zhao, Ahmed Saeed, Mostafa Ammar, Ellen Zegura

[18] **Characterizing the Effects of Rapid LTE Deployment: A Data Driven Analysis**
IEEE/IFIP TMA 2019
Kareem Abdullah, Noha Othman Korany, Ayman Khalafallah, Ahmed Saeed, Ayman Gaber

[19] **If you can't beat them, augment them: Improving Local WiFi with Only Above-driver Changes**
IEEE ICNP 2018
Ahmed Saeed, Mostafa Ammar, Ellen Zegura, Khaled Harras

[20] **Carousel: Scalable Traffic Shaping at End-Hosts**
ACM SIGCOMM 2017
Ahmed Saeed, Nandita Dukkipati, Vytautas Valancius, Vinh The Lam, Carlo Contavalli, Amin Vahdat

[21] **Argus: Realistic Target Coverage by Drones**
ACM/IEEE IPSN 2017
Ahmed Saeed, Ahmed Abdelkader, Mouhyemen Khan, Azin Neishaboori, Khaled Harras, Amr Mohamed
– Full version published in ACM Transactions on Sensor Networks 2019

[22] **Local and Low-cost White Space Detection**
IEEE ICDCS 2017
Ahmed Saeed, Khaled Harras, Ellen Zegura, Mostafa Ammar

[23] **The Inapproximability of Illuminating Polygons by α-Floodlights**
CCCG 2015
Ahmed Abdelkader, Ahmed Saeed, Khaled Harras, Amr Mohamed

[24] **Up and Away: A Visually-Controlled Easy-to-Deploy Wireless UAV Cyber-Physical Testbed**
IEEE WiMob 2014
Ahmed Saeed, Azin Neishaboori, Khaled Harras, Amr Mohamed

[25] **Low complexity target coverage heuristics using mobile cameras** *(short paper)*
IEEE MASS 2014
Azin Neishaboori, Ahmed Saeed, Khaled Harras, Amr Mohamed

[26] **On Target Coverage in Mobile Visual Sensor Networks**
ACM MOBIWAC 2014
Azin Neishaboori, Ahmed Saeed, Khaled Harras, Amr Mohamed

[27] **Location-Aware Probabilistic Route Discovery for Cognitive Radio Networks**
IEEE WCNC 2014
Ahmed Elbagori, Ahmed Saeed, Moustafa Youssef

[28] **A Low-Cost Large-Scale Framework for Cognitive Radio Routing Protocols Testing**
IEEE ICC 2013
Ahmed Saeed, Mohamed Ibrahim, Khaled Harras, Moustafa Youssef

[29] **RASID: A Robust WLAN Device-free Passive Motion Detection System**
IEEE PerCom 2012
Ahmed E. Kosba, Ahmed Saeed, Moustafa Youssef
– Full version published in IEEE Journal of Selected Topics in Signal Processing 2014

Journal Publications

[1] **On Realistic Target Coverage by Autonomous Drones**
ACM Transactions on Sensor Networks 2019
Ahmed Saeed, Ahmed Abdelkader, Mouhyemen Khan, Azin Neishaboori,
Khaled Harras, Amr Mohamed

[2] **Toward Dynamic Real-time Geo-location Databases for TV white Spaces**
IEEE Network 2015
Ahmed Saeed, Mohamed Ibrahim, Khaled Harras, Moustafa Youssef

[3] **Ichnaea: A Low-overhead Robust WLAN Device-free Passive Localization System**
IEEE Journal of Selected Topics in Signal Processing 2014
Ahmed Saeed, Ahmed E. Kosba, Moustafa Youssef

[4] **Nuzzer: A Large-Scale Device-Free Passive Localization System for Wireless Environments**
IEEE Transactions on Mobile Computing 2013
Moustafa Seifeldin, Ahmed Saeed, Ahmed E. Kosba, Moustafa Youssef, Amr El-Keyi

Workshop Publications

[1] **Modeling the Interactions between Core Allocation and Overload Control in μs-Scale Network Stacks**
IEEE MASCOTS Workshop
Jehad Hussien, Pratyush Sahu, Eric Stuhr, Ahmed Saeed

[2] **Understanding QUIC's Throughput Speedbumps**
ANRW 2025
Saubhik Mukherjee, Demi Lei, Mostafa Ammar, Ahmed Saeed

[3] **Lightweight Automated Reasoning for Network Architectures**
HotNets 2024
Rahul Bothra, Akshay Narayan, Venkat Arun, Ahmed Saeed, Brighten Godfrey

[4] **Do We Need a Million Satellites in Orbit?**
LEO-NET 2024 in conjunction with ACM MobiCom
Demi Lei, Ahmed Saeed

[5] **Astrolabe: Modeling RTT Variability in LEO Networks**
LEO-NET 2023 in conjunction with ACM MobiCom
Vaibhav Bhosale, Ketan Bhardwaj, Ahmed Saeed

[6] **Vision: The Case for Symbiosis in the Internet of Things**
ACM MCS 2015 in conjunction with ACM MobiCom
Ahmed Saeed, Mostafa Ammar, Khaled Harras, Ellen Zegura

[7] **Towards a Characterization of White Spaces Databases Errors: An Empirical Study**
ACM WiNTECH 2014 in conjunction with ACM MobiCom
Ahmed Saeed, Khaled Harras, Moustafa Youssef

[8] **Target Coverage Heuristics Using Mobile Cameras**
Robotic Sensor Networks 2014 part of Cyber-Physical Systems Week
Azin Neishaboori, Ahmed Saeed, Amr Mohamed, Khaled Harras

Refereed Posters and Demos

[1] **When Server-Level Load Imbalance Helps: Power Savings without Capacity Loss**
USENIX NSDI 2026 (Poster)
Sherif Abdelrazek, Dhyey Thummar, Ahmed Saeed

[2] **Understanding Interactions between Overload Control Core Allocation in Low-Latency Network Stacks**
ACM SIGCOMM 2023 (Poster)
Eric Stuhr, Ahmed Saeed

[3] **A Robust Technique for WLAN Device-free Passive Motion Detection**
ACM MobiCom 2011 (Poster)
Ahmed E. Kosba, Ahmed Saeed, Moustafa Youssef

[4] **Quantifying the Impact of GPU Specific Optimizations: An Experimental Study on a Weather Forecasting Application**
ACM PACT 2010 (Poster)
Ahmed Saeed, Emad Elwany, Emad Tawadros, Kareem Abdelsalam, Pakinam Yousry, Samia Hafez

[5] **DNIS: A Middleware for Dynamic Multiple Network Interface Scheduling**
ACM MobiCom 2009 (Poster)
Ahmed Saeed, Karim Habak, Mahmoud Fouad, Moustafa Youssef

[6] **DNIS: Dynamic Multiple Network Interface Scheduler**
ACM MobiCom 2009 (Demo)
Ahmed Saeed, Karim Habak, Mahmoud Fouad, Moustafa Youssef

Patents

US Patent 10,454,835
"Device And Method For Scalable Traffic Shaping With A Time-Indexed Data Structure"
Carlo Contavalli, Nandita Dukkipati, *Ahmed Issa*, Vytautas Valancius. Granted Oct. 2019.

US Patent 9,800,946
"Dynamic real-time TV white space awareness"
Khaled Harras, Moustafa Amin Youssef, Mohammed Ibrahim, *Ahmed Issa*. Granted Oct. 2017.

Service

Leadership
PC Co-Chair, IFIP Networking 2025.

Program Committees

2027	NSDI
2026	NSDI; CLOUD (Senior PC Member); LEO-NET
2025	CoNEXT; LEO-NET
2024	CoNEXT; INFOCOM; APSys; LEO-NET
2023	NSDI; ANRW; SysDW; LEO-NET
2022	SIGCOMM; HotNets; ANRW; APNet
2021	ANCS

External Reviewing
IEEE/ACM Transactions on Networking (ToN), IEEE Journal on Selected Areas in Communications (JSAC), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Mobile Computing (TMC), IEEE Computer, IEEE Networking Letters, ACM Transactions on Sensor Networks (TOSN), Elsevier Pervasive and Mobile Computing, Elsevier Ad Hoc Networks, Elsevier Computer Networks, IEEE VTC2015-Spring.

Georgia Tech Service

2026	SCS Chair Search Committee
2024-2026	Networking Area Coordinator
2022-2025	Member, CoC Communications Faculty Advisory Board
2022-2024	Member, PhD Visit Day Committee
2022	Member, PhD Admissions Committee

Talks

Demystifying Datacenters: AI – Energy – Sustainability

BBISS Insight Series	Feb. 2026

Fantastic Congestion Points and Where to Find Them

Alexandria University	Dec. 2024
Applied Innovation Center in Alexandria, Egypt	Dec. 2024
American University in Cairo	Nov. 2024
Texas Tech University	Mar. 2023

Improving the Performance and Resilience of LEO Satellite Networks

DARPA Forward at Georgia Tech	Oct. 2022

High Performance Network Stacks for Edge Servers

Google Networking Research Summit 2022	Feb. 2022

Building Scalable Network Stacks for Modern Applications

University of Texas at Austin	Apr. 2021
École Polytechnique Fédérale de Lausanne (EPFL)	Apr. 2021
University of Toronto	Mar. 2021
Microsoft Research	Mar. 2021
Imperial College London	Mar. 2021
George Mason University	Mar. 2021
Simon Fraser University	Mar. 2021
University of British Columbia	Mar. 2021
Virginia Tech	Feb. 2021
University of Waterloo	Feb. 2021
Max Planck Institute - SWS	Feb. 2021
Georgia Tech	Feb. 2021
King Abdullah University of Science and Technology (KAUST)	Jan. 2021

Scalable Packet Scheduling in Software

Massachusetts Institute of Technology	Feb. 2019
Princeton University	Feb. 2019
University of Southern California	Oct. 2018
University of California - Santa Cruz	Oct. 2018